Importance Sampling

English translation of the Polish original

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Introduction to IS

Importance Sampling (IS) is a simulation technique used to estimate rare events.

Objective:

- Classical Monte Carlo (MC) needs a very large sample to estimate rare-event probabilities precisely.
- IS reduces the number of samples needed for the same accuracy.

How does Importance Sampling work?

The main idea of IS is:

1. Change the probability distribution of the random variable X to a *biasing distribution* $f^*(x)$.
2. Weight samples from $f^*(x)$ using the weight function $W(x)$:

$$W(x) = \frac{f(x)}{f^*(x)},$$

where $f(x)$ is the original distribution.

Estimation: IS estimates the event probability $P(X \geq t)$ as:

$$P(X \geq t) = \mathbb{E}^* \left[W(X) \cdot \mathbb{I}(X \geq t) \right],$$

where $\mathbb{E}^*$ denotes expectation under $f^*(x)$.

Advantages of IS:

- Focuses simulations on *important* regions where rare events occur.
- Reduces estimator variance compared with classical MC.
- Uses a **biasing distribution** that increases the frequency of rare events.

Example:

- Classical MC: few samples satisfy $X \geq t$, leading to high variance.
- IS: with a suitable $f^*(x)$, most samples satisfy $X \geq t$.

General definition of IS:

$$\mathbb{E}[g(X)] = \int g(x)f(x)dx = \int g(x)W(x)f^*(x)dx = \mathbb{E}^*[g(X)W(X)],$$

where:

- $g(X)$ is the function of the random variable whose expectation is estimated,
- $f(x)$ is the original probability distribution,
- $f^*(x)$ is the biasing distribution,
- $W(x) = \frac{f(x)}{f^*(x)}$ is the weight function.

Variance of the IS estimator:

$$\mathrm{Var}^*[g(X)W(X)] = \mathbb{E}^*[g^2(X)W^2(X)] - (\mathbb{E}^*[g(X)W(X)])^2 .$$

Distribution-changing techniques

Scaling and **Translation** are two basic ways to change distributions:

- **Scaling:** Rescale the random variable to concentrate probability mass in the region of interest.
- **Translation:** Shift the distribution to increase the number of samples in the rare-event region.

Scaling

Scaling rescales a random variable X by a factor $a > 1$:

$$X^* = aX.$$

Biasing distribution:

$$f^*(x) = \frac{1}{a} f\left(\frac{x}{a}\right),$$

where $f(x)$ is the original distribution of X.

Weighting: The weight function $W(x)$ is the density ratio:

$$W(x) = \frac{f(x)}{f^*(x)} = \frac{af(x)}{f(\frac{x}{a})}$$

Scaling example: Weibull distribution

Consider a random variable X with a Weibull distribution:

$$f(x) = \frac{b}{\lambda}\left(\frac{x}{\lambda}\right)^{b-1} e^{-(x/\lambda)^b}, \quad x \geq 0.$$

Biasing distribution after scaling:

$$f^*(x) = \frac{b}{a\lambda}\left(\frac{x}{a\lambda}\right)^{b-1} e^{-\left(\frac{x}{a\lambda}\right)^b}.$$

Weight function:

$$W(x) = a^b e^{-(1-a^{-b})(\frac{x}{\lambda})^b}$$

Translation

Translation shifts a random variable X by c:

$$X^* = X + c, \quad c > 0.$$

Biasing distribution:

$$f^*(x) = f(x - c),$$

where $f(x)$ is the original distribution of X.

Weighting: The weight function $W(x)$ is the density ratio:

$$W(x) = \frac{f(x)}{f^*(x)} = \frac{f(x)}{f(x - c)}.$$

Translation example: Gaussian distribution

Consider a random variable X with a normal distribution $N(0,1)$:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}.$$

Biasing distribution after translation:

$$f^*(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-c)^2}{2}}, \quad c > 0.$$

Weight function:

$$W(x) = e^{-\frac{c(2x-c)}{2}}.$$

Comparison of Scaling and Translation

- **Scaling:**
  - Effective for homogeneous variables, such as the Weibull distribution.
  - May spread probability mass into unwanted regions.
- **Translation:**
  - More effective for symmetric distributions, such as the Gaussian.
  - Avoids the "dimensionality effect" encountered in scaling.

Exponential Twisting

Exponential Twisting (ET) is an importance-sampling technique based on:

- changing the distribution of X by "twisting" its density,
- using a distribution with a modified moment-generating function.

Objective:

- ET minimizes estimator variance by optimally *redirecting probability mass* toward important-event regions.
- It is particularly effective for rare-event estimation, where classical approaches may fail.

Density twisting: Generate a new density $f^*(x)$ by twisting the original density $f(x)$:

$$f^*(x) = \frac{e^{sx} f(x)}{M(s)},$$

where:

- $M(s) = \mathbb{E}[e^{sX}]$ is the moment-generating function,
- $s > 0$ is the *twisting* parameter, optimized to minimize estimator variance.

Weighting: The weight function $W(x)$ for a sample x is:

$$W(x) = \frac{f(x)}{f^*(x)} = e^{-sx+\mu(s)},$$

where $\mu(s) = \log M(s)$.

Parameter s

Choose s to minimize the variance
of the IS estimator.

Optimality condition:

$$\mu'(s) = t,$$

where t is the rare-event threshold, for example in $P(X \geq t)$.

Interpretation: The optimal s *redirects probability mass* toward the region of interest, such as the distribution tail.

Example: Gaussian distribution

Consider $X \sim N(0,1)$. Its moment-generating function is:

$$M(s) = \exp\left(\frac{s^2}{2}\right).$$

Twisted density:

$$f^*(x) = \frac{e^{sx} f(x)}{M(s)} = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-s)^2}{2}}.$$

Weighting:

$$W(x) = e^{-sx+\frac{s^2}{2}}.$$

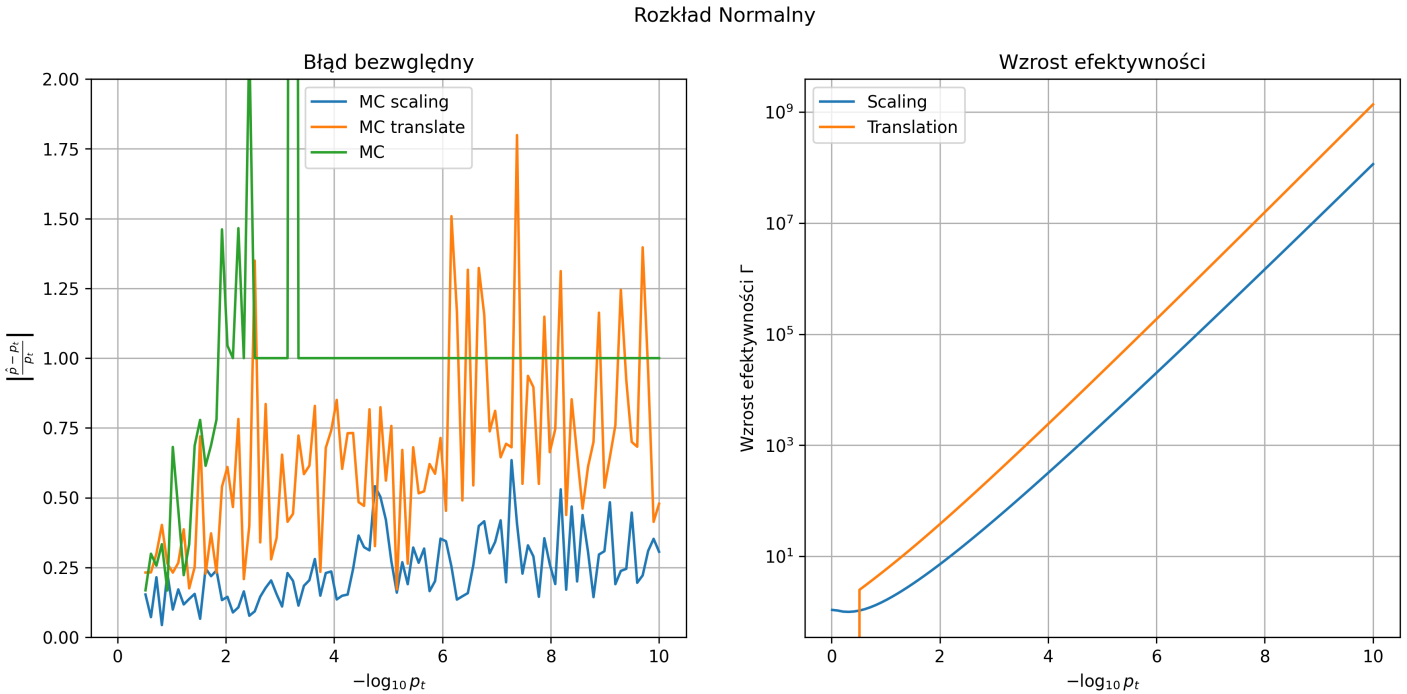
Advantages of Exponential Twisting

- **Efficiency:** Minimizes the variance of the IS estimator.
- **Applicability:** Can be used with normal, gamma and other distributions.
- Uses moment-generating functions known for many standard distributions.

NUMERICAL TESTING

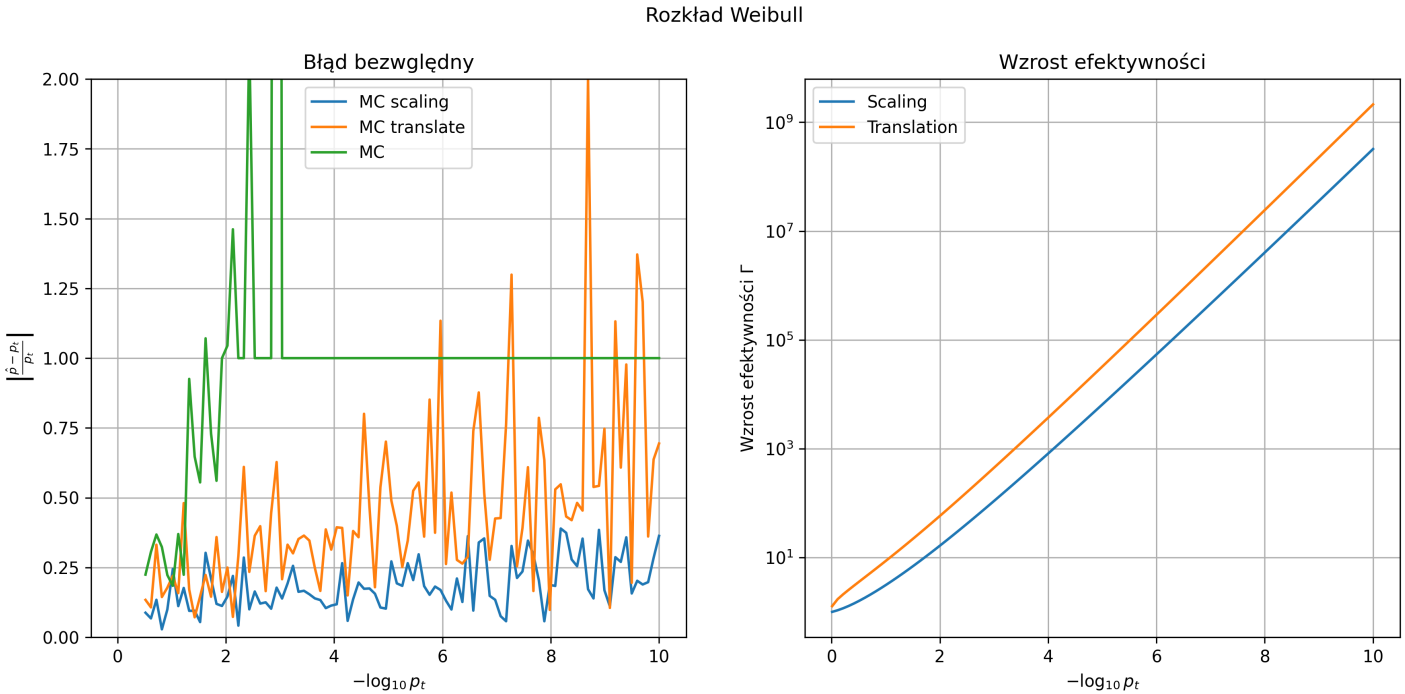
Numerical testing

Normal distribution



Numerical testing

Weibull distribution



References



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